Towards a Cartography of Datacentering

A Multimodal Research Agenda

Babajide Owoyele

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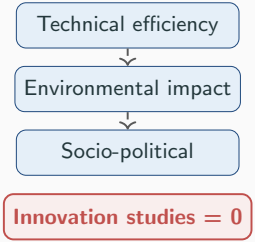
The Empirical Provocation

4,757 papers on data centers (OpenAlex, 2010–2025)

Growth: 372% in 15 years

Silo	Papers
Technical efficiency	≈ 279
Environmental impact	≈ 114
Socio-political	≈ 15
Innovation studies	0



What "The Cloud" Conceals

"Cloud" does ideological work:

- Naturalises infrastructure — makes it weightless, placeless
- Conceals: water, land, power, fibre, diesel generators
- $\approx$ 400 **TWh/yr** globally — fastest-growing OECD electricity user

Make it visible: actors, contestation, governance, regime change

Resource	Scale
Electricity	400 TWh/yr
Water	ML/day
Land	Industrial
Finance	REIT, PE, SWF

Introducing "Datacentering"

Concept

Datacentering = the active socio-material process through which digital infrastructure is continuously assembled, contested, and reproduced

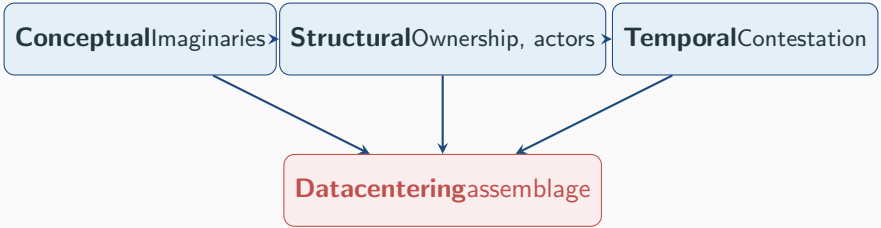
Why a gerund?

- Process over static object
- Who is doing the datacentering?
- What is assembled or foreclosed?

Theoretical grounding

- Orlikowski (2007): sociomaterial entanglement
- Jasanoff & Kim (2015): imaginaries co-produce facts
- "Cloud" suppresses what datacentering names

Theoretical Framework: CAS as Sensemaking Lens



After **Phillips & Ritala (2019)** — sensemaking lens, not a formal model

Data center assemblages = multi-actor, nonlinear, emergent

A Multimodal Research Agenda

WP	CAS Dimension	Data	Output
WP1	Conceptual	Operator websites, Wikipedia	Imaginary divergence (ECI)
WP2	Conceptual	Satellite + corporate imagery	Visual abstraction score
WP3	Structural	SEC EDGAR, OpenCorporates, REIT, Overture Maps	Ownership + urban graph
WP4	Temporal	News, planning, social media (GDELT)	Signal-to-response lag

Why multimodal?

Strategic opacity is a *sociotechnical fact* — not a data gap. A firm that won't publish is still in a

WP1 & WP2: The Conceptual Dimension

WP1 — Territorial Imaginaries

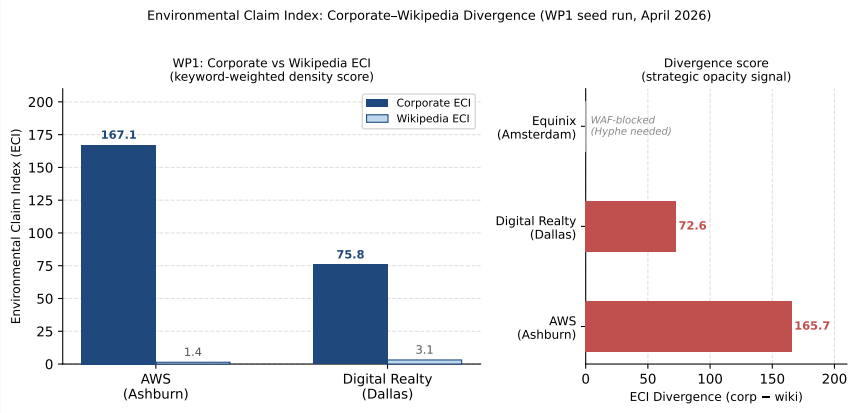
- Corporate websites vs. city Wikipedia
- ECI divergence score
- High sustainability / low community = placelessness production

WP2 — Strategic Invisibility

- Sentinel-2 + LST + VIIRS nightlights
- Corporate photography (CLIP)
- Visual abstraction score
- Seed: 3 facilities, GEE pipeline

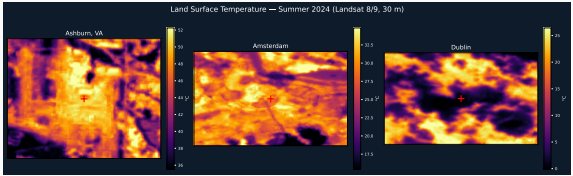
How do actors construct imaginaries that make datacentering **legible or illegible***?*

WP1 First Results: ECI Divergence (Seed Run, April 2026)

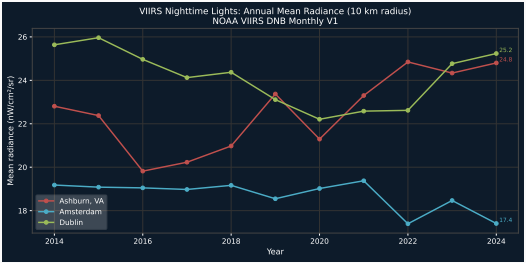


AWS ECI 120× higher than Wikipedia. Equinix WAF-blocked: strategic opacity at the *technical* layer.

WP2 First Results: Satellite Layers (Seed Run, April 2026)



LST thermal maps — summer 2024, 30 m (Landsat 8/9)



VIIRS nighttime lights 2014–2024

Ashburn: +8.7% — only net growth.

Amsterdam: −9.2% (LED transition effect).

WP3 & WP4: Structural and Temporal Dimensions

WP3 — Structural Cartography

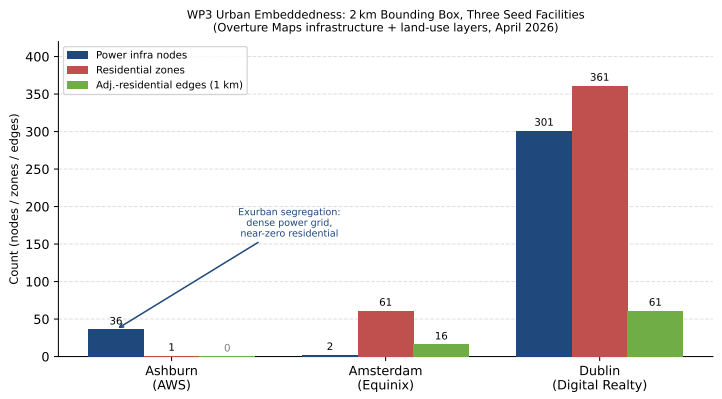
- SEC EDGAR + REIT filings: 20 nodes, 35 edges
- **REIT structure** disperses accountability
- Overture Maps + city2graph: urban graphs
- Ashburn: 36 power nodes, 1 res. zone

WP4 — Contestation Signals

- News (GDELT), planning records, social media
- BERTopic + event detection
- **Signal-to-response lag**
- Seed: 58 events, 4 cities

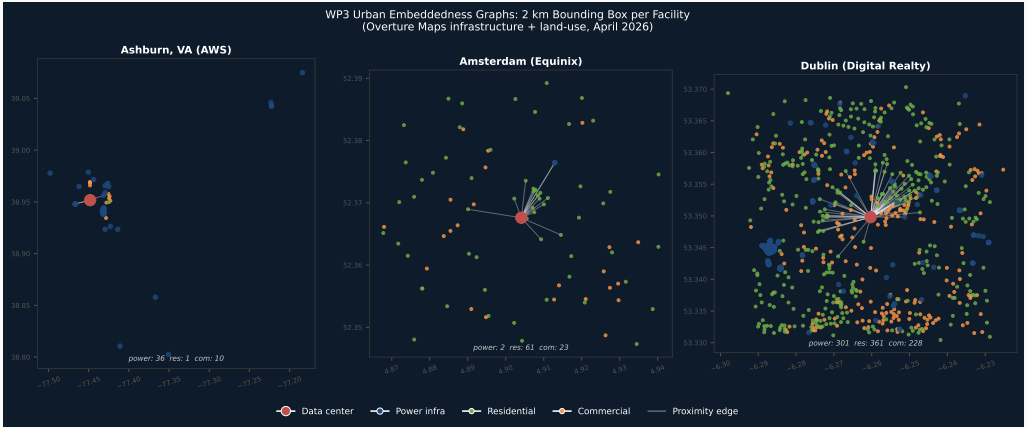
Who does the datacentering, in what legal form, and what happens when communities resist?

WP3 First Results: Urban Embeddedness (Seed Run, April 2026)



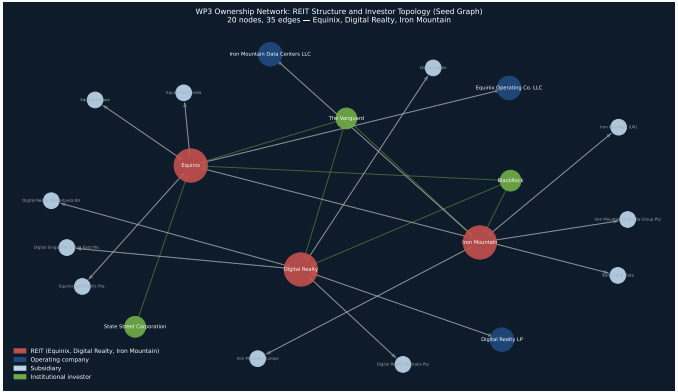
Ashburn: near-zero residential adjacency despite dense power grid. Amsterdam/Dublin: 16–61 adjacency edges — embedded in the urban fabric.

WP3: Urban Graphs as Spatial Evidence



• data center • power infra • residential lines = proximity edges

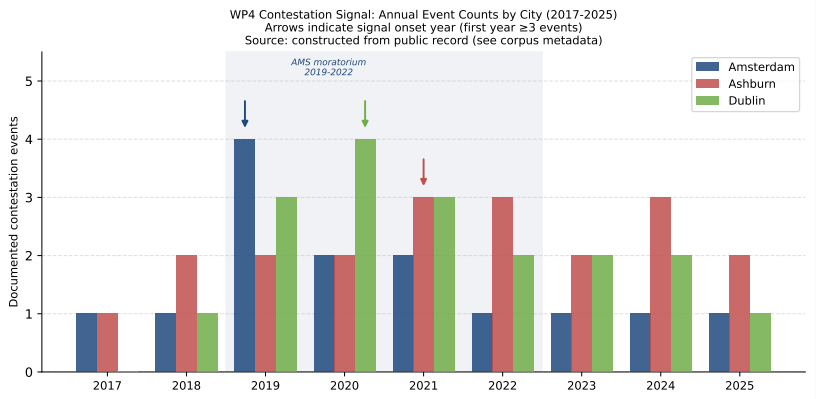
WP3b First Results: Ownership Network (Seed Run, April 2026)



• REIT • operating • subsidiary • institutional investor

Vanguard, BlackRock, State Street hold cross-REIT positions: no single accountable owner.

WP4 First Results: Contestation Signal (Seed Corpus, April 2026)



Shaded band = Amsterdam moratorium (2019–2022). Dublin: 81% negative-tone events — highest sustained signal. Ashburn: comparable volume, zero regime response.

Illustrative Comparison: Virginia vs Amsterdam

Dimension	Northern Virginia	Amsterdam
Ownership	REIT-dominated, dispersed	Mixed; operator-owned, AMS-IX coop
Counterparty	None identifiable	Municipality + named operators
Urban (2 km)	36 power / 1 res. zone	2 power / 61 res. zones
ECI divergence	+165.7 (corp 120× wiki)	Lower; regulatory vocab absorbed
Contestation	High	High
Regime response	**None**	**Moratorium 2019–2022**
Key variable	Not signal strength — ownership structure + governance counterparty	

Building the Cartography Together

Why open exchange is a theoretical necessity

Datacentering is CAS-like: too large, too distributed, too opaque for any single team. The research commons is the methodological response.

What we share:

- Community facility index (CC BY 4.0)
- NLP / CV / network pipelines (MIT)
- Open data and open workflows

Concrete asks:

- Local knowledge: disputes, unreported facilities
- Company registry or REIT data access
- Methodological critique
- Replication in other languages

The cloud is not a metaphor. It is infrastructure.
And infrastructure can be mapped.